



Digital savings and financing household emergencies: what is the reality in Sub-Saharan Africa?

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Abstract

Digital savings now fulfil the monetary functions of transaction and precaution, enabling households to cope with everyday shocks such as education and health emergencies. The impact of digital savings on emergency financing can be attributed to a number of factors. Firstly, there is a reduction in borrowing, transfer and financial transaction costs. Secondly, there is the removal of liquidity, time and mobility constraints that can arise in emergency situations. Thirdly, there is the adoption of digital savings tools that facilitate their effective use. The objective of this study is to analyse the effects of digital savings on the financing of education and health emergencies for households in SSA. The data for the years 2021, 2017, 2014 and 2011 for Sub-Saharan African (SSA) countries was obtained from the World Bank's Global Findex and World Development Indicators. The estimation method is based on a fractional Probit with instrumental variables to correct for endogeneity. The findings indicate that the vast majority of digital savings indicators exert a favourable and substantial influence on emergency financing. Furthermore, the impact of precautionary digital savings indicators is found to be more significant than that of transactional digital savings. In addition, the development of mechanisms that limit the use of digital savings through competition, such as insurance and easy access to credit, has been observed. This recommendation is twofold: firstly, it calls for the enhancement of households' financial capacities and purchasing power, thereby enabling them to accumulate precautionary digital savings; secondly, it advocates the implementation of policies that promote, disseminate and facilitate the utilisation of digital savings tools.

Keywords Digital savings · Emergencies · Precautionary savings · Transaction savings

JEL Classification D14 · D03 · G00 · G21 · R51

1 Introduction

The advent of digitalisation in developing countries, particularly on the African continent, has been identified as a primary catalyst for the observed surge in financial inclusion over the past decade. In sub-Saharan Africa (SSA), for instance, 42.6% of the adult population had a bank account in 2017, compared to only 23.2% in 2011, and nearly 21% of adults had a mobile bank account (Global Findex 2022). This trend signifies a distinctive opportunity for financial service providers (FSPs) to more effectively fulfil their mission and expand their customer base. It has been demonstrated by numerous studies that households and businesses with access to financial services exhibit greater resilience to financial shocks in comparison to those lacking such access (Moore et al. 2019). This phenomenon has been observed in Chile, where low-income women who were members of microfinance institutions and had received free savings accounts were able to reduce their dependence on credit and improve their ability to finance emergencies during economic crises (Pomeranz and Kast 2022).

In the period between 2017 and 2021, the average account ownership rate in developing economies demonstrated an increase of 8 percentage points, with the proportion of adults in possession of an account rising from 63 to 71%. Furthermore, the proportion of adults who make or receive digital payments increased from 35% in 2014 to 57% in 2021. Furthermore, it was found that nearly two-thirds of digital payment recipients also used their accounts for cash management purposes; approximately 40% used their accounts for savings; and 40% of payment recipients formally took out a loan. In the case of SSA, this increase is largely attributable to the adoption of mobile money. The Global Findex report (2021) also highlights that digital savings represent the most reliable source of emergency financing. However, as Global Findex data from 2021 indicates, 30% of adults in developing economies rely on informal sources, such as friends and family, to obtain money in an emergency. The report also indicates that women are more likely than men to encounter difficulties in accessing emergency financing. Indeed, a survey revealed that 50% of women in developing economies reported that they could easily obtain money in an emergency within 30 days, compared to 59% of men.

By the close of 2020, the SSA had accounted for 64% of the value of mobile money transactions on a global scale, reaching USD 490 billion in 2020, representing an increase of 23% in comparison with the previous year. The region accounts for 53% of all monthly active mobile money accounts. Across the SSA region, there was an approximate 21% increase in transaction volumes, reaching approximately 45 trillion between 2021 and 2022. In East Africa, transaction volumes exhibited an 18% year-on-year increase, while in West Africa, volumes surged by nearly a third, reaching approximately 12 billion in 2022. While the value of transactions in East Africa increased by 23% year-on-year, growth in West Africa was 22% (rising from USD 239 billion in 2021 to USD 277 billion in 2022).

Digital financial services such as mobile money facilitate the secure storage of funds and enable the efficient transfer of funds over long distances at a low cost, thereby increasing remittances and consumption and thereby boosting investment (Global Findex 2021; Nzepang and Nguenda Anya 2023). The facilitation of pay-

ments has been demonstrated to encourage the wider adoption of financial services, as it is more straightforward to retain funds in an account until a need for expenditure arises and then to make a direct payment. Conversely, once funds are deposited into an account, it becomes more straightforward to retain them for the purpose of savings. Digital savings can thus be categorised into two distinct roles: transaction savings and precautionary savings. However, the receipt of a payment into an account, particularly if it can be utilised to substantiate a consistent income stream over an extended period, can concurrently streamline the formal borrowing process.

The originality of this study lies in two points. On the one hand, it distinguishes between two types of digital savings: precautionary digital savings and transactional digital savings. This distinction highlights the function of digital savings as a payment and precautionary instrument. Secondly, this study focuses on the financing of households' daily emergencies through digital savings. This differs from work on financial inclusion, which generally associates financial inclusion with well-being, autonomy, education and health, but neglects the financing of households' daily emergencies. The objective of this study is to analyse the effects of digital savings on the financing of emergencies for households in SSA. This study is important for at least three reasons. First, it contributes to a better alignment between digital savings and insecurity in the daily, family and social lives of households with a view to reducing the level of difficulties related to education and health. Second, this study highlights digital savings fulfilling the monetary function of transaction and precaution. Finally, this study enriches the literature on the importance of digital financial inclusion for household well-being in particular, and economic and social development in general.

The remainder of this paper is organised as follows: Sect. 2 reviews the literature, Sect. 3 presents the methodology, Sect. 4 presents and discusses the results of the analysis, and Sect. 5 concludes.

2 Literature review

2.1 Theoretical review

In recent years, there has been a consistent and sustained increase in digital savings, with ASS being the primary catalyst for this growth. The extant literature on the effects of digital savings on household emergency financing is based on three theories: constraint theory (Goldratt and Cox 1984), transaction cost theory (Coase 1937; Williamson 1990) and innovation appropriation theory (Certeau 1980). Firstly, as per the tenets of constraint theory, the accumulation of digital savings empowers households to effectively manage constraints pertaining to liquidity, temporal constraints, mobility-related expenses, financial institutions, and other costs that emerge in emergency circumstances. Consequently, digital savings can be utilised as a means to overcome various constraints in emergency situations. Secondly, as posited by transaction cost theory, the advent of digital savings has led to a reduction in the financial transaction costs and an enhancement in the ease of execution. Consequently, financing emergencies through digital savings has become a more cost-effective alternative to traditional methods such as borrowing or seeking aid. Finally, innovation appro-

priation theory assumes that mastery of digital tools facilitates the use of digital savings to finance emergencies.

The advent of digital savings represents a recent phenomenon, and academic research in this domain remains in its infancy (Kauffman and Riggins 2012). The paucity of research in this area is evident in the dearth of theoretical and empirical studies addressing this issue. The majority of the work is comprised of prospective analyses of the challenges posed by ICTs through the prism of digital household savings. The contributions of Kauffman and Riggins (2012), Weber et al. (2012) and Riggins and Weber (2013) are among the few studies on the link between digital savings and economic development. In their 2012 publication, Kauffman and Riggins engage in a discursive analysis of the role and impact of ICTs at the consumer, micro-finance and foreign investor levels. In line with this, the present study hypothesises that digital savings facilitate the financing of household emergencies in sub-Saharan Africa, by positively affecting the probability of financing educational and medical emergencies.

2.2 Empirical review

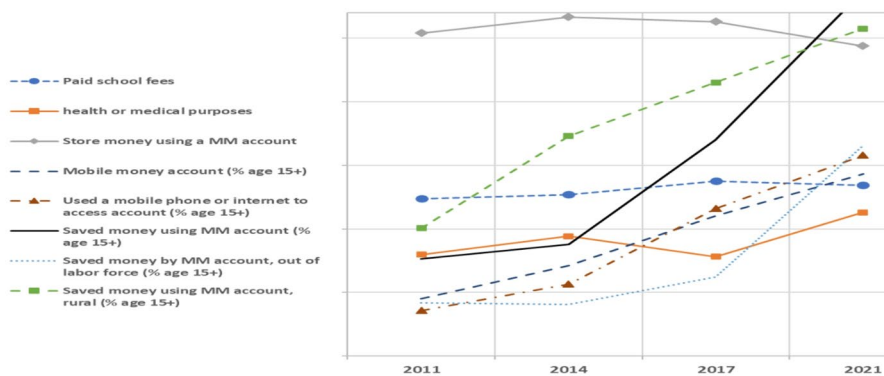
Empirical studies illustrate these dynamics and shows that mobile money enables households to mobilize distant social networks, smooth consumption during income shocks, and reduce dependence on informal credit (Jack and Suri 2014; Lee et al. 2021a, b). Moreover, commitment savings and digital wage programs enhance women's financial autonomy and household welfare, while evidence from multiple Sub-Saharan African countries confirms the positive impact of digital savings on resilience to income, health, and climate-related shocks (Ashraf et al. 2010; Field et al. 2021a, b; Batista and Vicente 2021; Abiona and Koppensteiner 2022; Kyagara 2025; Nzepang et al. 2024; Onuegbu et al. 2025). These studies collectively illustrate a progression from descriptive observations to robust evidence on how digital savings strengthen household financial resilience.

The integration of digital payments into wages and public transfers has further expanded the role of digital savings in emergency management. Mobile wage payments increase saving behaviour and reduce informal withdrawal costs, while digital transfers lower administrative expenses and enhance efficiency (Blumenstock et al. 2018; Breza et al. 2020; Aker et al. 2016; Muralidharan et al. 2016a, b). Shocks such as the COVID-19 pandemic have highlighted the capacity of digital accounts to serve as rapid channels for emergency support and liquidity management (Gentilini et al. 2020a, b; World Bank 2021; IMF 2023; GSMA 2025). Despite these advances, the literature remains dominated by single-country analyses, with cross-country comparisons and the effects of systemic shocks still underexplored. Indeed, the diffusion of digital savings and household behaviour with regard to financial innovations is not uniform across SSA countries. This study addresses these gaps by examining how households across multiple Sub-Saharan African countries and distinguishes between two types of digital savings who are precautionary digital savings and transactional digital savings. This paper mobilizes digital savings to respond to emergencies, contributing to a more comprehensive understanding of the dynamics of digital financial

inclusion. Figure 1 below illustrates the evolution of digital savings and emergency financing indicators in SSA.

Figure 1 shows a marked increase in the adoption and use of mobile money in sub-Saharan Africa between 2011 and 2021. Indicators related to mobile money accounts, digital savings and the use of telephones to access financial services are all trending upwards, albeit at varying rates. The opening of mobile money accounts is progressing steadily, accompanied by a notable increase in savings made through these accounts, particularly after 2017, when the curve shows a clear acceleration. Savings in rural areas are following a similar trajectory, but at a higher level, suggesting that mobile money plays a crucial role in rural financial inclusion. Emergency-related uses, such as the payment of medical or school fees, are also increasing, but at a more moderate and sometimes fluctuating rate, indicating that these services are being adopted gradually and remain sensitive to economic conditions, while the storage of money via a mobile account has been on a downward trend since 2014.

However, households' ability to finance emergencies depends on other demographic, economic, infrastructural and financial factors in a country. For example, access to electricity facilitates the use of digital services to quickly transfer and access the funds needed during crises (Aker and Mbiti 2010; Dinkelman 2011; Ntieche and Nzepang 2025). Similarly, if natural resource rents are well managed, they can be an additional source of financing, although their volatility may limit their effectiveness (Auty 2001; Collier and Goderis 2012). In addition, high income and a large proportion of adults increase savings capacity and access to insurance or micro-savings instruments, strengthening resilience to shocks (Modigliani and Brumberg 1954; Deaton 1992). Conversely, a high dependency ratio and high final consumption can reduce households' financial margin and limit their ability to mobilise resources for emergencies (Bloom et al. 2003; Carroll 1997). Finally, access to domestic credit allows households to finance emergencies without liquidating their assets, increasing their financial flexibility (Stiglitz and Weiss 1981; Beck et al. 2007; Karlan and Zinman 2010). Specifically in relation to health emergencies, studies on financial resilience also highlight that households employ strategies combining savings, social



Sources: Authors, with Global Index data

Fig. 1 Evolution of emergencies and digital savings in SSA

support and access to credit to cope with unexpected expenses, particularly for health emergencies (BMC Public Health 2025; Ishiwatari and Sasaki 2024).

3 Methodology

The first sub-section presents the data and variables used, while the second focuses on the specification of the model and the estimation method.

3.1 Data and variables

The data utilised in this study is drawn from Global Findex, a comprehensive source of data on global access to financial services, encompassing payments, savings and borrowing. The Global Findex is a comprehensive survey that provides nearly 300 indicators on a range of financial topics, including account ownership, payments, savings, credit and financial resilience. The data is presented for all indicators by country, region and income group, offering a detailed and nuanced understanding of global financial habits and resilience. All available indicators are presented for the years 2021, 2017, 2014 and 2011. It should be noted, that this study focuses exclusively on data relating to sub-Saharan African countries and thus draws on data from 42 countries in this region (Appendix 1). Furthermore, the data are also summarised by gender, income level (adults living in the richest 60% and poorest 40% of households), labour force participation (working and non-working adults), age (young people, adults and older people), and rural or urban residence. It is true that the use of multiple independent cross-sectional data, known as the pseudo-panel method, can dilute the reality at the household level. However, Nobel Prize winner Deaton (1985) was the first to confirm the usefulness of this method for studies of household behaviour and several other studies use this method for behavioural studies (Guillerm 2017).

With regard to the variables under consideration, the household urgency is measured by two indicators: the proportion of individuals who set aside funds for financing education and health-related emergencies (Lee et al. 2021a, b). In relation to digital savings indicators, the present study differentiates between three transaction indicators and three precautionary indicators. The transaction indicators under consideration are as follows: the proportion of individuals who have a mobile money account, who temporarily hold cash via mobile money, and who use a telephone or the internet to access their mobile money account. Precautionary indicators include: the proportion of individuals who save money in mobile money accounts firstly, the proportion of adults in rural areas who save money in mobile money accounts secondly, and the proportion of individuals outside the labour force who save money in mobile money accounts finally. As demonstrated in the extant literature, the financing of emergencies can be explained by several factors, including per capita income (GDP per capita), the level of household expenditure, access to electricity, the adult population, the old-age dependency ratio, natural resource rents, domestic credit granted to the private sector by banks, the percentage of arable land, and the CPIA gender equality index (Wresinski 1987; Parizot et al. 2002; Field et al. 2021a, b). The

latter variables are derived from the World Bank's World Development Indicators and are all expressed as percentages.

3.2 Econometric specification and estimation method

The objective of this paper is to measure the effects of digital savings on the financing of emergencies for households in sub-Saharan Africa. Given that the variables measuring emergencies are the proportion of individuals who set aside funds for financing education and health-related emergencies are proportions, a fractional Probit panel model is utilised for their modelling (Papke and Wooldridge 2008). Indeed, the application of linear regression models would not be an appropriate estimation technique when the dependent variable is a fraction or a proportion. The fractional Probit model, as introduced by Papke and Wooldridge (1996), is utilised as a relevant estimator due to its capacity to handle variables such as rates, proportions and fractional data. The fractional Probit model is a fractional response model for a dependent variable that is greater than or equal to 0 and less than or equal to 1 (Papke and Wooldridge 1996). The fractional Probit model is quasilielihood estimators and a model of the mean of the dependent variable y conditional on covariates X , which we denote by μ_X . Because y is in $[0; 1]$, we must ensure that μ_X is also in $[0; 1]$. We do this in this paper by using a heteroskedastic probit model for μ_X . The key insight from quasilielihood estimation is that you do not need to know the true distribution of the entire model to obtain consistent parameter estimates. In fact, the only information that we need is the correct specification of the conditional mean (Papke and Wooldridge 2008).

As a model, our dependent variable Y , $0 \leq Y \leq 1$ is to be explained by a $1 * K$ vector of explanatory variables $K = (x_1, x_2 \dots x_K)$, with the convention that $x_1 = 1$. The population model is:

$$E(Y|X) = \beta_1 + \beta_2 x_2 + \dots + \beta_K x_K = X\beta \quad (1)$$

If Y is strictly between zero and one then a linear model for the log-odds ratio is:

$$E\left(\log\left[\frac{Y}{1-Y}\right] | X\right) = X\beta \quad (2)$$

The precariousness equations are modelled as follow:

$$Y_{it}^j = \beta E_num_{it}^j + \theta_k X_{it}^j + \varepsilon_{it} \quad (3)$$

With $j = 1, 2$ for education and health. i is the country indicator and t is the period indicator, which is the year in this study. Y_{it}^j is therefore the urgency j of country i at time t . E_num is the proportion of digital savings used, X_{it}^j is the vector of control variables and ε_{it} is the error term. β and θ_k are the model parameters. However, there is a problem of endogeneity of digital savings on household urgency.

It is a commonly held view that impoverished households are marginalised within the formal banking system and that digital savings represent their sole means of saving. Conversely, endogeneity may also arise from measurement errors (false declarations about emergencies or digital savings). In such circumstances, a rudimentary fractional Probit model is inadequate for the estimation of the causal effects of digital savings on household emergencies. The utilisation of a fractional Probit model with instrumental variables is imperative (Papke and Wooldridge 1996).

The challenge posed by an instrumental variable model is to identify instruments that are correlated with digital savings but not with emergencies (Cole et al. 2018). The prevailing approach in microeconomic literature is to utilise the number of mobile phone agents present within the local vicinity (Koomson et al. 2021). The hypothesis is that the number of mobile phone agents in the neighbourhood has a direct relationship with the adoption of digital savings, but not directly with emergencies. The presence of a mobile money agent in the neighbourhood functions as a conduit for cash withdrawal, catering to both individuals with and without bank accounts. The extant microeconomic literature employs either the distance to the nearest mobile money point of sale (Jack and Suri 2014) or the time taken to reach the nearest point of sale (Afawubo et al. 2020). These two variables influence the use of mobile money, yet do not directly affect urgency (Mvogo and Djouaka 2022).

In accordance with the macroeconomic approach adopted in this study, Two instruments are used: mobile phone penetration rates and the number of secure internet servers in 2010, 2013, 2016 and 2020 for data from 2011, 2014, 2017 and 2021 respectively, both of which are available on the World Development Indicators website. Indeed, it is highly likely that the levels of mobile phones and secure Internet servers in 2010 explain the growth in the number of mobile phone agents in the country, and therefore in close proximity to individuals within the population in subsequent years, without directly explaining the financing of emergencies. Thus, the levels of mobile phones and the number of secure internet servers in 2010 would act as lagged variables generally used in macroeconomic studies. This finding suggests a direct correlation with the adoption of digital savings, but not with emergency situations.

4 Results

We present, on the one hand, the results of descriptive analyses and, on the other, the results of econometric analyses.

4.1 Descriptive analysis results

Table 1 below sets out the descriptive indicators in Box A, providing an overview of the major structural and behavioural trends in SSA. Panel B focuses exclusively on the correlation matrix.

This table shows in panel (A) that the average proportion of adults (aged 15 and over) seeking emergency financial resources to pay school fees is 26.1%, with a standard deviation of 8.1%, which highlights the financial pressures faced by families and

the importance of education as a critical item of household expenditure, consistent with Goldratt's (1984) constraint theory, which posits that households must allocate limited resources to essential needs, and with Jensen and Meckling's (1976). In terms of health emergencies, 18.3% of adults borrowed urgently for health reasons, ranging from 0.7% to 40.8% (with a standard deviation of 7.6%). This last result shows marked differences in health coverage or access to medical care. With regard to the digital savings variables, it is noted that the average percentage of people who store money via a mobile account is 51.4%, with a significant range from 15.6% to 75.8%. This finding suggests a considerable adoption of mobile technology as a financial intermediation tool, and considerable heterogeneity in this practice across the region. Considering the percentage of adults who have actually saved via a mobile account, the average rate drops to 8%, which is well below the usage rate for simple storage. This discrepancy indicates that mobile money is predominantly utilised as a transfer or temporary deposit mechanism, rather than as a formal savings instrument, in line with the findings of Jack and Suri (2014), Lee et al. (2021a, b), and Suri (2021). This phenomenon may be attributed to a variety of factors, including irregular income, immediate consumption needs, or a lack of awareness regarding the benefits of saving. This phenomenon is particularly evident among adults outside the labour force, of whom only 4.5% utilise mobile savings platforms. This phenomenon is also evident among rural populations, where an average rate of 9.3% has been observed, thereby substantiating the notion that remote areas and marginalised populations continue to be underrepresented in digital savings channels. This underrepresentation highlighting the combined effects of limited access, financial literacy, and structural disparities on digital financial inclusion (Aron 2023; Kyagara 2025; Certeau 1980). With regard to mobile money account ownership, the average is 18.5%, with a relatively high standard deviation of approximately 17.3%. A similar dynamic can be observed for access to financial accounts via mobile or internet, with an average rate of 18.6% and a standard deviation of 15.2%. This underscores the heterogeneity among nations with regard to technological integration and financial inclusion.

With regard to the remaining variables, the mean adult population per nation is 14 million, whilst the mean rate of electricity access is a mere 42.4%, thus emphasising one of the primary impediments to economic advancement and the utilisation of information technology. Furthermore, the average demographic dependency ratio of 82.6% underscores the pressure exerted on the working population by young people and the elderly. This demographic burden can present a challenge for savings and investment policies, thereby reducing households' capacity to accumulate capital. Another salient statistic is the average final consumption rate relative to GDP, which stands at 83.5%, thus highlighting the absence of investment orientation in several economies. In a similar vein, the credit allocated to the private sector remains modest, with an average share of 19.6% of GDP. This is further compounded by the average percentage of arable land, which stands at 16.5%. Consequently, inadequate levels of financial intermediation are compounded by the prevailing agricultural potential. Furthermore, average per capita income is USD 2,135, ranging from USD 203 to over USD 10,841, which demonstrates significant variability. Finally, with regard to technological indicators (mobile subscriptions, internet usage, secure servers), the average figure is almost 80 mobile subscriptions per 100 inhabitants, while the aver-

Table 1 Descriptive statistics and pairwise correlations

(A) Descriptive Statistics of variables																			
	Variable	Mean	Std. Dev	Min	Max														
(1)	Urgent for Paid school fees purposes (% age 15+)	26.1	08.1	06.6	51.3														
(2)	Urgent borrowed for health or medical purposes (% age 15+)	18.3	07.6	0.70	40.8														
(3)	Store money using a mobile money account (% with a mobile money account)	51.4	09.8	15.6	75.8														
(4)	Saved money using a mobile money account (% age 15+)	08	07.4	0.70	37.2														
(5)	Saved money using a mobile money account, out of labor force (% age 15+)	04.5	04.8	0.10	23.5														
(6)	Saved money using a mobile money account, rural (% age 15+)	09.3	06.4	0.40	33.5														
(7)	Mobile money account (% age 15+)	18.5	17.3	00	72.9														
(8)	Used a mobile phone or the internet to access an account (% age 15+)	18.6	15.2	0.40	71.8														
(9)	Population adult (%)	14,251,175	19,570,668	406,616.34	1.165e+08														
(10)	Access to electricity (% of population)	42.372	24.573	4.1	99.6														
(11)	Age dependency ratio (% of working-age population)	82.6	14.164	38.407	107.749														
(12)	Total natural resources rents (% of GDP)	11.387	9.661	0.003	51.844														
(13)	GDP per capita (current US\$)	2135.947	2459.191	230.071	10,841.685														
(14)	Final consumption expenditure (% of GDP)	83.546	16.834	34.778	129.965														
(15)	Domestic credit to private sector by banks (% of GDP)	19.604	17.306	1.691	98.954														
(16)	Arable land (% of land area)	16.529	14.44	0.437	52.25														
(17)	CPIA gender equality rating (1 = low to 6 =high)	3.302	0.538	01.5	04.5														
(18)	Mobile cellular subscriptions (per 100 people)	79.683	36.367	20.248	168.924														
(19)	Individuals using the internet (% of population)	19.63	17.884	0.9	78.7														
(20)	Secure internet servers (per 1 million people)	219.891	1564.073	0.071	16,057.58														
(B) Pairwise correlations																			
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)
1																			
0.02	1																		
0.12	-0.03	1																	
0.23 ×	0.17 ×	0.27 ×	1																
0.24 ×	0.14*	0.27 ×	0.78+	1															

Table 1 (continued)

(B) Pairwise correlations																				
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)	(17)	(18)	(19)	(20)	
(6)	0.44+	0.048	0.31+	0.67+	0.67+	1														
(7)	0.42+	0.18 ×	0.23 ×	0.70+	0.66+	0.70+	1													
(8)	0.42+	0.04	0.16*	0.53+	0.52+	0.67+	0.83+	1												
(9)	0.12	-0.14*	0.03	0.01	-0.07	0.05	-0.03	-0.06	1											
(10)	0.33+	-0.33+	0.08	0.23 ×	0.26 ×	0.35+	0.34+	0.46+	0.07	1										
(11)	-0.34+	0.31+	-0.08	-0.22 ×	-0.24 ×	-0.33+	-0.26 ×	-0.47+	0.08	-0.76+	1									
(12)	-0.21 ×	0.09	-0.17 ×	-0.06	-0.06	-0.26 ×	-0.19 ×	-0.30+	-0.01	-0.26 ×	0.29+	1								
(13)	0.28 ×	-0.40+	0.08	0.01	0.11	0.18 ×	0.12	0.31+	-0.08	0.72+	-0.77+	-0.09	1							
(14)	-0.10	0.13	-0.00	-0.01	-0.06	0.01	-0.08	-0.01	0.10	-0.09	0.02	-0.43+	-0.30+	1						
(15)	0.27 ×	-0.33+	0.09	0.12	0.18 ×	0.31+	0.19 ×	0.47+	-0.07	0.58+	-0.72+	0.70+	0.11	1						
(16)	-0.04	-0.06	0.15*	0.03	0.02	0.07	-0.06	0.06	0.15*	0.02	-0.06	-0.30+	-0.15*	0.34+	0.10	1				
(17)	0.26 ×	-0.25 ×	0.18 ×	0.20 ×	0.15*	0.22 ×	0.26 ×	0.34+	-0.15*	0.35+	-0.54+	-0.39+	0.38+	0.07	0.48+	0.18 ×	1			
(18)	0.25 ×	-0.29+	0.08	0.22 ×	0.22+	0.30+	0.37+	0.45+	-0.09	0.78+	-0.66+	-0.24 ×	0.67+	-0.22 ×	0.54+	-0.12	0.45+	1		
(19)	0.31+	-0.13	0.09	0.43+	0.42+	0.45+	0.53+	0.61+	0.05	0.77+	-0.70+	-0.27 ×	0.62+	-0.06	0.54+	-0.05	0.39+	0.76+	1	
(20)	0.15*	-0.11	-0.15*	0.04	-0.02	-0.04	0.09	0.18 ×	0.16*	0.25 ×	-0.32+	-0.09	0.26 ×	-0.01	0.32+	-0.04	0.14*	0.30+	0.36+	1

Sources: Authors, with Global FinIndex data

* $p < 0.1$; × $p < 0.05$; + $p < 0.01$

age number of secure servers is 220 per million. However, the mean internet user rate is 19.63%, with a range between 17.8 and 78.7%, thus highlighting the very uneven nature of internet penetration in SSA.

In the quadrant (B) which focuses on correlations, an initial observation is that emergency spending on education is not correlated with emergency spending on health. However, it is important to note that each of these two variables is generally significantly correlated with the three variables of mobile money savings, mobile money account ownership, and access to financial accounts via mobile or internet. Moreover, an absence of correlation has been demonstrated between emergency spending on education and health and the utilisation of mobile financial services for money storage. Conversely, the variables illustrating the percentage of adults who have actually saved via a mobile account, mobile money account ownership, and access to financial accounts via mobile or internet are strongly correlated, indicating that mobile account ownership logically goes hand in hand with increasing use for savings or digital payments. The close links observed may be indicative of the fact that mobile money services are not merely a solitary use case, but rather, are utilised by individuals for a diverse range of applications, including storage, transfer, and consumption.

With regard to other variables, it is observed that access to electricity has a significant positive correlation with several variables related to mobile money, thus revealing the major structural effect of electrification. Furthermore, an association has been demonstrated between electricity and demographic dependency, suggesting that nations with higher levels of electrification can support a greater demographic burden while maintaining a level of infrastructure development. In a similar vein, access to electricity has been shown to be strongly correlated with GDP per capita, Internet usage and mobile subscriptions. This highlights the central role of income in the development of basic infrastructure and the adoption of digital technologies. Conversely, GDP exhibits a negligible correlation with mobile savings, suggesting that an increase in income does not necessarily result in a corresponding rise in digital savings rates. This phenomenon may be attributable to the presence of behavioural, technological or institutional impediments that hinder the utilisation of mobile savings.

4.2 Econometric results

We present the basic results of our study and the robustness analyses of these results. Before proceeding, it is necessary to check for an important econometric problem, namely multicollinearity between explanatory variables. Table 1 shows that the risk of multicollinearity between explanatory variables is low. Indeed, Gujarati (2003) mentions that the problem of multicollinearity can be declared when the coefficients exceed 0.80, whereas none of our correlation coefficients reach 0.5. Thus, we can assume that multicollinearity between variables is not a problem in the context of this study. However, we perform Theil's (1971) variance inflation factor (VIF) test in Table 2 below.

Table 2 shows that the VIF values for all independent variables are less than 10. A model does not suffer from multicollinearity for VIF values < 10.

Table 2 Variance inflation factor (VIF) test

	(1)	(2)	(3)	(4)	(5)	(6)
Store money using a mobile money account (% with a mobile money account)	3.15					
Saved money using a mobile money account (% age 15+)		4				
Saved money using a mobile money account, out of labor force (% age 15+)			2.08			
Saved money using a mobile money account, rural (% age 15+)				3.09		
Mobile money account (% age 15+)					1.93	
Used a mobile phone or the internet to access an account (% age 15+)						3.05
Log Population adult (%)	2.53	2.66	1.83	2.44	4	4.66
Access to electricity (% of population)	3.14	2.81	3.56	3	4.31	2.17
Age dependency ratio (% of working-age population)	2.96	1.58	0.98	2.38	3.34	3
Total natural resources rents (% of GDP)	2	2.7	2.03	2.49	1.07	2.9
GDP per capita (current US\$)	3.29	2	2.19	2.05	2.21	1.06
Final consumption expenditure (% of GDP)	1.99	1.95	2.02	2.77	3.18	1
Domestic credit to private sector by banks (% of GDP)	2.23	3.75	1.86	1.09	2	1
Arable land (% of land area)	3.36	4	2.39	3	2	3.52
CPIA gender equality rating (1=low to 6=high)	1.48	4.11	1.99	4.06	3.88	2.59
Average	2.613	2.956	2.093	2.637	2.792	2.495

Sources: Authors, with Global Findex data

4.2.1 Basic econometric results

Table 3 below shows the marginal effects of the basic results obtained using the fractional Probit method. The regressions are statistically robust and significant, with Wald probabilities consistently below 0. The Hansen tests for instrument exogeneity validity confirm that the instruments are appropriate, supporting the reliability of the estimations for assessing the impact of digital savings on emergency financing. Indeed, Hansen's testing principle is that instruments are valid if the probability of the test is above 10%, which is the case for all our probabilities.

With regard to the impact of digital savings on emergency financing, the findings indicate that all digital savings indicators exhibit a positive and significant effect on the financing of education and health emergencies in SSA, with the exception of temporary deposits, which are deemed to be non-significant. For example, when savings through a mobile money account increase by 1%, the probability of financing school emergencies increases by 0.151, while the probability of financing medical emergencies increases by 0.263. These results are consistent with the findings of Jack and Suri (2014), who highlighted the central role of mobile money in Kenya in households' financial resilience to income shocks. The findings of this study demonstrate that the majority of households utilise digital savings as a means of financing unanticipated expenses. This phenomenon can be explained by the fact that digital savings enable households to overcome several constraints in emergency situations and make financial transactions less costly, according to constraint and transaction cost theories (Goldratt and Cox 1984; Williamson 1990). Digital savings therefore enable house-

Table 3 Marginal effects of digital savings on emergency finance

Dependent variable: school purposes	(1)	(2)	(3)	(4)	(5)	(6)
Store money using a mobile money account (% with a mobile money account)	0.048 (0.42)					
Saved money using a mobile money account (% age 15+)		0.151* (0.07)				
Saved money using a mobile money account, out of labor force (% age 15+)			0.248** (0.02)			
Saved money using a mobile money account, rural (% age 15+)				0.411*** (0.00)		
Mobile money account (% age 15+)					0.144*** (0.00)	
Used a mobile phone or the internet to access an account (% age 15+)						0.150*** (0.00)
Log Population adult (%)	0.007*** (0.01)	0.002*** (0.01)	0.003*** (0.01)	0.001*** (0.01)	0.001*** (0.00)	0.003*** (0.00)
Access to electricity (% of population)	0.000 (0.32)	0.000 (0.60)	0.000 (0.56)	0.000 (0.88)	-0.000 (0.82)	0.000 (0.86)
Age dependency ratio (% of working-age population)	-0.001 (0.30)	-0.001 (0.46)	-0.001 (0.41)	-0.001 (0.51)	-0.001 (0.38)	-0.001 (0.48)
Total natural resources rents (% of GDP)	-0.011** (0.03)	-0.002** (0.02)	-0.002** (0.02)	-0.004 (0.11)	-0.009* (0.08)	-0.012* (0.07)
GDP per capita (current US\$)	-0.000 (0.34)	-0.000 (0.92)	-0.000 (0.75)	0.000 (0.72)	0.000 (0.59)	0.000 (0.84)
Final consumption expenditure (% of GDP)	-0.022** (0.02)	-0.011** (0.04)	-0.009** (0.04)	-0.021 (0.10)	-0.017 (0.15)	-0.015 (0.10)
Domestic credit to private sector by banks (% of GDP)	0.000 (0.54)	0.000 (0.69)	0.000 (0.74)	-0.000 (0.78)	0.000 (0.89)	-0.000 (0.61)
Arable land (% of land area)	-0.001 (0.22)	-0.001 (0.30)	-0.001 (0.27)	-0.001 (0.31)	-0.000 (0.56)	-0.001 (0.33)
CPIA gender equality rating (1 = low to 6 = high)	0.017 (0.20)	0.015 (0.26)	0.017 (0.17)	0.017 (0.16)	0.009 (0.44)	0.014 (0.26)
<i>Instrumental equation</i>						
Mobile cellular subscriptions (per 100 people)	0.025**	0.041**	0.030**	0.015*	0.013***	0.013*

Table 3 (continued)

Dependent variable: school purposes	(1)	(2)	(3)	(4)	(5)	(6)
	(0.02)	(0.02)	(0.02)	(0.08)	(0.00)	(0.06)
Individuals using the internet (% of population)	0.021	0.033***	0.012***	0.022***	0.016***	0.026***
	(0.45)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Secure internet servers (per 1 million people)	0.002***	0.001***	0.001**	0.005***	0.003**	0.006**
	(0.00)	(0.00)	(0.01)	(0.00)	(0.05)	(0.04)
Country-specific effects	Yes	Yes	Yes	Yes	Yes	Yes
Years-specific effects	Yes	Yes	Yes	Yes	Yes	Yes
Hansen test for instrument exogeneity	0.182	0.233	0.147	0.451	0.286	0.312
N	141	141	141	141	141	141
Wald chi2(22)	78.77	175.82	155.75	279.20	203.44	189.71
Wald probability	0.000	0.000	0.000	0.000	0.000	0.000
Dependent variable: Health or medical purposes	(1)	(2)	(3)	(4)	(5)	(6)
Store money using a mobile money account (% with a mobile money account)	0.037					
	(0.63)					
Saved money using a mobile money account (% age 15+)		0.263***				
		(0.00)				
Saved money using a mobile money account, out of labor force (% age 15+)			0.364***			
			(0.00)			
Saved money using a mobile money account, rural (% age 15+)				0.247**		
				(0.01)		
Mobile money account (% age 15+)					0.160***	
					(0.00)	
Used a mobile phone or the internet to access an account (% age 15+)						0.170***
						(0.00)
Log Population adult (%)	0.008**	0.005**	0.001**	0.007**	0.001**	0.004**
	(0.02)	(0.02)	(0.03)	(0.02)	(0.02)	(0.03)
Access to electricity (% of population)	-0.000	-0.000	-0.000	-0.000	-0.001**	-0.001
	(0.61)	(0.16)	(0.21)	(0.26)	(0.04)	(0.11)
Age dependency ratio (% of working-age population)	-0.001	-0.000	-0.000	-0.000	-0.000	-0.000
	(0.38)	(0.79)	(0.62)	(0.57)	(0.54)	(0.68)
Total natural resources rents (% of GDP)	0.000	-0.000	-0.000	0.000	0.000	0.000

Table 3 (continued)

Dependent variable: Health or medical purposes	(1)	(2)	(3)	(4)	(5)	(6)
	(0.87)	(0.83)	(0.94)	(0.69)	(0.63)	(0.64)
GDP per capita (current US\$)	-0.011**	-0.004	-0.008*	-0.002*	-0.011	-0.014
	(0.01)	(0.25)	(0.10)	(0.10)	(0.31)	(0.15)
Final consumption expenditure (% of GDP)	0.000	0.001	0.001	0.001	0.001*	0.001*
	(0.33)	(0.16)	(0.16)	(0.16)	(0.06)	(0.09)
Domestic credit to private sector by banks (% of GDP)	-0.011	-0.013	-0.021	-0.014	-0.001	-0.016**
	(0.37)	(0.23)	(0.24)	(0.18)	(0.11)	(0.04)
Arable land (% of land area)	-0.000	-0.000	-0.000	-0.000	-0.000	-0.000
	(0.30)	(0.45)	(0.38)	(0.40)	(0.77)	(0.41)
CPIA gender equality rating (1 = low to 6 = high)	-0.019	-0.024*	-0.020*	-0.019	-0.027**	-0.022*
	(0.12)	(0.05)	(0.10)	(0.11)	(0.02)	(0.06)
<i>Instrumental equation</i>						
Mobile cellular subscriptions (per 100 people)	0.018**	0.014**	0.016**	0.009***	0.018*	0.011**
	(0.04)	(0.02)	(0.02)	(0.01)	(0.09)	(0.04)
Individuals using the internet (% of population)	0.021*	0.023***	0.012***	0.012***	0.026***	0.021***
	(0.05)	(0.00)	(0.00)	(0.00)	(0.00)	(0.00)
Secure internet servers (per 1 million people)	0.001***	0.000***	0.004**	0.003***	0.011**	0.007**
	(0.00)	(0.00)	(0.01)	(0.00)	(0.05)	(0.05)
Country-specific effects	Yes	Yes	Yes	Yes	Yes	Yes
Years-specific effects	Yes	Yes	Yes	Yes	Yes	Yes
Hansen test for instrument exogeneity	0.308	0.177	0.527	0.190	0.444	0.143
N	141	141	141	141	141	141
Wald chi2(22)	108.26	217.09	154.17	155.46	237.19	255.33
Wald probability	0.000	0.000	0.000	0.000	0.000	0.00

Sources: Authors, with Global Findex data

p-value in parenthesis ()

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$.

holds to finance emergencies by quickly restructuring their demand for transactional or precautionary money. Moreover, the findings indicate that the impact of precautionary digital savings indicators is more pronounced than that of transactional digital savings. This phenomenon can be attributed to the inherent uncertainty surrounding emergency situations, which necessitate that households develop the capacity to anticipate such events. However, given the instability of digital transaction savings, it is the digital precautionary savings that can effectively finance emergencies.

A further salient finding is that the proportion of individuals outside the labour force who have saved through a mobile account increases the probability of financing educational emergencies by 0.248 and the probability of medical emergencies by 0.364. This finding lends further support to Certeau's theory of innovation appro-

priation (1980) and demonstrates that demographic groups historically marginalised by conventional economic structures (i.e., homemakers, young people who have exited formal education, military personnel stationed in barracks, and retirees) can utilise digital savings as a tool for financial empowerment, thereby enhancing their resilience in managing unanticipated circumstances. Thus, these marginalised individuals, who still lack the physical and financial means for mobility and communication due to their standard of living and occupation, find in digital savings a way to restructure their savings. This finding is consistent with the conclusions of Field et al. (2021a, b), who demonstrated the empowering effect of digital payments on women in India. Moreover, the finding that the proportion of individuals residing in rural areas who have saved through a mobile account increases the probability of financing school emergencies by 0.411 and the probability of medical emergencies by 0.247 underscores the significance of digital savings in contexts characterised by low banking density, where traditional financial institutions are frequently absent or challenging to access. This additional finding lends further support to the assertions put forth by transaction cost theory, which posits that digital technologies serve to reduce the financial costs associated with accessing financial services and to enable arbitrage in emergency circumstances. Finally, the use of a mobile phone or the internet to access an account has been shown to significantly enhance emergency financing capabilities. This finding serves to corroborate the theory of technological appropriation, thereby demonstrating that a high level of proficiency and familiarity with digital instruments are pivotal in determining the intensity of their utilisation and the efficacy with which they function as financial management instruments.

In addition to the variables pertaining to digital savings, other factors have been identified as playing a significant role in determining the capacity of households to manage emergency financing. For instance, the size of the adult population (logarithm) has been demonstrated to increase the probability of financing educational emergencies in all specifications, as well as that of financing medical emergencies. Indeed, an increase in the size of the adult population could lead to a growth in the number of digital savings users. Consequently, this could result in an enhancement of intra-family solidarity with regard to education and health, facilitated by these digital savings. In addition, it is evident that the final consumption percentage in relation to GDP has a mitigating effect on the likelihood of financing school emergencies. This phenomenon can be attributed to the observation that, in contexts where current consumption absorbs a significant proportion of national income, savings margins are constrained, thereby limiting households' capacity to accumulate reserves for use in times of emergency. This finding aligns with the conclusions of Prina (2015) in Kenya, who underscored the significance of incentive programmes in enhancing savings among economically disadvantaged populations. Conversely, access to private bank credit, measured by the share of private sector credit in GDP, which generally exerts negative and insignificant effects, nevertheless significantly reduces the likelihood of financing medical emergencies when using a mobile phone or the internet to access an account. This result may be indicative of a substitution or competition effect, suggesting that households utilise digital savings to a lesser extent in economies where bank credit is more readily available. Another interesting finding concerns the gender equality index (CPIA gender equality rating) and GDP per capita, which has no sig-

nificant effect on funding for educational emergencies but generally has a significant negative effect on funding for medical emergencies. This counterintuitive result may suggest that higher GDP and gender equality are associated with a lower probability of relying on savings or emergency finance mechanisms to meet healthcare expenses. Thus, these countries could have alternative institutional mechanisms that limit the use of digital savings, such as health insurance, targeted subsidies and easy access to bank credit. This result is consistent with life cycle theory and the permanent income hypothesis, which suggest that households with higher and more stable incomes are better able to smooth their consumption and build up financial reserves *ex ante*. In higher-income countries, the existence of more developed social protection systems, formal health insurance mechanisms and greater access to traditional financial services may reduce the need to resort to emergency solutions, including through digital finance. Similarly, according to institutional economics theories, improved gender equality is generally accompanied by better access for women to education, formal employment and social protection systems, which reduces their financial vulnerability to health shocks. Finally, the share of natural resource rents in GDP reduces the likelihood of financing school emergencies. This could be interpreted in light of the paradox of abundance, according to which natural resource wealth does not necessarily translate into improved living conditions for households. This can limit individuals' ability to meet school expenses.

4.2.2 Robustness analysis

To test the robustness of our estimates, we carry out three analyses. The first analysis, presented in Table 4, is based on countries' income levels, while the second analysis, presented in Table 5, is based on regional economic communities. With regard to analyses by regional economic communities, we mainly distinguish between the Southern African Development Community (SADC), the Common Market for Eastern and Southern Africa (COMESA), the East African Community (EAC), the Economic Community of West African States (ECOWAS) and the Economic Community of Central African States (ECCAS). Finally, Table 6 presents the estimates obtained using two other methods: fractional Logit with instrumental variables and Tobit with instrumental variables.

This table illustrates that, in general, digital savings have a substantial impact on the allocation of funding for school emergencies, as the level of development of a nation increases. Indeed, it is widely accepted that the level of development is inextricably linked to the development of digital infrastructure. To illustrate this point, consider the example of lower-middle-income countries, where the marginal amplitude associated with saving via a mobile money account reaches 0.188, in contrast to 0.311 in upper-middle-income countries. This outcome lends further support to transaction cost theory, which posits that economic agents are more inclined to adopt digital solutions due to the reduction of structural impediments such as banking and technological networks. This finding is consistent with the work of Kauffman and Riggins (2012), who emphasised the pivotal role of technological infrastructure in the efficacy of digital financial services. However, we encounter situations where the magnitude of the marginal effect diminishes, becomes negative and becomes non-

Table 4 Analysis by income level

	LI	LMI	UMI	LI	LMI	UMI
	School purposes			Health or medical purposes		
Store money using a mobile money account (% with a mobile money account)	0.021	0.008	-0.103	0.028	-0.030	0.021
	(0.84)	(0.92)	(0.34)	(0.82)	(0.78)	(0.79)
Saved money using a mobile money account (% age 15+)	-0.094	0.188*	0.311*	0.246	0.226**	-0.130
	(0.59)	(0.09)	(0.06)	(0.10)	(0.02)	(0.18)
Saved money using a mobile money account, out of labor force (% age 15+)	0.015	0.248	0.533**	0.379**	0.243	0.163
	(0.94)	(0.17)	(0.03)	(0.04)	(0.13)	(0.28)
Saved money using a mobile money account, rural (% age 15+)	0.314*	0.539***	0.031	0.370**	0.155	-
	(0.09)	(0.00)	(0.89)	(0.03)	(0.30)	(0.04)
Mobile money account (% age 15+)	0.060	0.174***	0.176***	0.184***	0.078	0.119**
	(0.44)	(0.00)	(0.00)	(0.00)	(0.14)	(0.03)
Used a mobile phone or the internet to access an account (% age 15+)	0.002	0.282***	0.138**	0.206***	0.029	-0.003
	(0.98)	(0.00)	(0.01)	(0.01)	(0.69)	(0.96)
<i>N</i>	66	56	19	66	56	19
Country-specific effects	Yes	Yes	Yes	Yes	Yes	Yes
Years-specific effects	Yes	Yes	Yes	Yes	Yes	Yes
Hansen test for instrument exogeneity	0.146	0.348	0.140	0.227	0.313	0.214

LI Low income, *LMI* Lower middle income, *UMI* Upper middle income

Sources: Authors, with Global Findex Data

p-value in parenthesis ()

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

significant for upper-middle-income countries. This finding underscores the potential for developed countries to have mechanisms in place that regulate the utilisation of digital savings. A parallel tendency is discernible in the realm of financing medical exigencies, thereby substantiating the observations of Field et al. (2021a, b) that the digitalisation of cash flows bestows upon women a heightened agency in decision-making, notably with respect to medical expenditures.

However, it is imperative to direct particular attention to individuals residing in rural areas. In lower-middle-income countries, the marginal effect for financing school expenses is 0.539, while for medical emergencies it is 0.370. These results, which are among the highest recorded, underscore the pivotal function of digital savings as a conduit for inclusivity among rural households, which frequently encounter challenges in accessing conventional financial services. This finding lends further support to the arguments of Williamson (1990) and Coase (1937) regarding the reduction of transaction costs through technology, and also corroborates the empirical conclusions of Prina (2015), who demonstrated that access to a savings account signifi-

Table 5 Analysis by regional economic community

	EAC	ECOWAS	ECCAS	COMESA	SADC
<i>Health or medical purposes</i>					
Store money using a mobile money account (% with a mobile money account)	0.433 (0.11)	-0.108 (0.38)	0.107 (0.60)	0.294*** (0.00)	0.056 (0.54)
Saved money using a mobile money account (% age 15+)	0.411 (0.12)	0.233*** (0.00)	0.245** (0.02)	0.334*** (0.01)	0.176 (0.28)
Saved money using a mobile money account, out of labor force (% age 15+)	0.259 (0.60)	0.442*** (0.00)	0.167 (0.31)	0.391*** (0.01)	0.400* (0.06)
Saved money using a mobile money account, rural (% age 15+)	1.025** (0.01)	0.430*** (0.00)	0.286 (0.12)	0.343** (0.01)	-0.076 (0.69)
Mobile money account (% age 15+)	0.346* (0.05)	0.156*** (0.01)	0.118* (0.08)	0.179*** (0.00)	0.083 (0.25)
Used a mobile phone or the internet to access an account (% age 15+)	0.053 (0.81)	0.197*** (0.01)	0.185 (0.13)	0.119 (0.13)	0.045 (0.53)
<i>School purposes</i>					
Store money using a mobile money account (% with a mobile money account)	0.110 (0.45)	0.040 (0.64)	-0.027 (0.91)	0.075 (0.50)	0.156* (0.05)
Saved money using a mobile money account (% age 15+)	-0.088 (0.79)	0.058 (0.64)	-0.019 (0.93)	0.158 (0.16)	-0.113 (0.32)
Saved money using a mobile money account, out of labor force (% age 15+)	-0.082 (0.84)	0.195 (0.26)	0.281 (0.29)	0.206 (0.18)	0.057 (0.64)
Saved money using a mobile money account, rural (% age 15+)	0.680** (0.02)	0.598*** (0.01)	0.200 (0.49)	0.348*** (0.01)	0.081 (0.66)
Mobile money account (% age 15+)	0.290** (0.02)	0.068 (0.35)	0.032 (0.74)	0.134** (0.01)	0.072 (0.12)
Used a mobile phone or the internet to access an account (% age 15+)	0.281 (0.12)	0.080 (0.42)	0.026 (0.86)	0.146** (0.04)	0.045 (0.35)
N	24	48	29	47	50

Sources: Authors, with Global Findex data

p-value in parenthesis ()

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

cantly increased food consumption in rural households headed by women. Moreover, in low-income countries, the overall marginal effects of technological access (i.e. the use of the telephone or internet to access an account) remain negligible. This situation is consistent with the argument posited in the literature review, which asserts that impediments pertaining to infrastructure, digital literacy and network coverage impede the adoption of digital savings (Grossman and Tarazi 2014). In summary, as demonstrated in Table 3, the impact of digital savings on emergency financing is

Table 6 Results using fractional Logit Tobit methods with instrumental variables

Dependent variable: School purposes	(1)	(2)	(3)	(4)	(5)	(6)
Store money using a mobile money account (% with a mobile money account)	0.021					
		(0.551)				
Saved money using a mobile money account (% age 15+)		0.119**				
		(0.03)				
Saved money using a mobile money account, out of labor force (% age 15+)			0.197***			
			(0.00)			
Saved money using a mobile money account, rural (% age 15+)				0.201***		
				(0.00)		
Mobile money account (% age 15+)					0.088**	
					(0.02)	
Used a mobile phone or the internet to access an account (% age 15+)						0.111***
						(0.00)
Country-specific effects	Yes	Yes	Yes	Yes	Yes	Yes
Years-specific effects	Yes	Yes	Yes	Yes	Yes	Yes
Hansen test for instrument exogeneity	0.398	0.118	0.155	0.144	0.210	0.522
N	141	141	141	141	141	141
Wald chi2(22)	124.02	199.88	105.33	89.30	366.01	173.44
Wald probability	0.000	0.000	0.000	0.000	0.000	0.000

Sources: Authors, with Global Findex data

p-value in parenthesis ()

* $p < 0.1$; ** $p < 0.05$; *** $p < 0.01$

found to be contingent upon a country's income level. It is evident that as a nation's technological and institutional environment evolves, the potential for digital savings to foster financial inclusion and resilience increases concomitantly.

With regard to the financing of medical emergencies, the COMESA community has the highest rate of money stored in mobile accounts. In the same region, other forms of digital savings, such as those made via accounts in rural areas or by people outside the labour force, serve to reinforce this trend, thereby highlighting the accessibility of digital savings tools in this region. Furthermore, the EAC is another region where digital savings appear to play a significant role in financing health emergencies. In the context of rural areas, the practice of saving is associated with a marginal amplitude of 1.025, representing the most significant impact observed in the entire table. It is important to note, however, that the majority of EAC countries are members of COMESA and therefore may adopt similar infrastructure policies. In contrast, the ECOWAS States and the ECCAS States have exhibited favourable and substantial outcomes from digital savings, though to a more modest degree than observed in the COMESA or the EAC. In the case of the SADC, the marginal effects appear to be generally more modest, or even absent. This finding suggests that digital savings do not have a significant impact on this region. This may be indicative of heterogeneity

in access to digital financial services, despite the presence of a country such as South Africa, which is equipped with advanced banking systems.

Analogous trends are emerging in the context of school emergencies. In the EAC, for instance, digital savings in rural areas exhibit a marginal amplitude of 0.680, in comparison to 0.598 in the ECOWAS and 0.348 in the COMESA. In a similar vein, the proportion of individuals in possession of a mobile money account exerts a marginal effect of 0.29 in EAC, as opposed to 0.134 in COMESA. Conversely, SADC and ECCAS demonstrate an almost total absence of significant effects of digital savings on the financing of school emergencies, with the exception of a marginally significant positive effect of -0.156 from those who temporarily store money via mobile money to settle day-to-day transactions. In conclusion, the effects of digital savings on emergency financing vary significantly across regional economic communities in SSA. While blocs such as the EAC and COMESA demonstrate robust results, indicating widespread and functional adoption, others such as SADC and ECCAS exhibit more equivocal performance.

These results shows that the effects of digital savings on the financing of medical and educational emergencies vary according to the income level of countries and the territorial and institutional characteristics specific to each region. Thus, digital savings have a more significant impact on financing educational emergencies in middle-income countries and a more significant impact on financing medical emergencies in low-income countries. However, these effects are mitigated with development, where alternative mechanisms or regulations exist. At the regional level, some regions experience more significant effects from digital savings, reflecting greater efforts and investments in the use and construction of digital infrastructure.

The results in Table 6 are similar to those in Table 3. Indeed, we also observe that all digital savings indicators have a positive and significant effect on financing educational and health emergencies in sub-Saharan Africa, with the exception of temporary deposits. This reinforces the findings that digital savings enable households to finance emergencies by quickly restructuring their demand for transactional or precautionary money. Similarly, we find that the impact of precautionary digital savings indicators is more pronounced than that of transactional digital savings.

5 Conclusion

The objective of this study is to analyse the effects of digital savings on the financing of household emergencies in SSA. Indeed, digital savings are now emerging as a primary instrument for financial inclusion on the African continent. Digital savings enable individuals to store funds securely and at minimal cost, and to transfer them expeditiously and economically over long distances. These funds can thus be regarded as transaction savings and precautionary savings, given the ease with which they can be retained in an account until the need for their expenditure arises, at which

point a direct payment can be made. In this sense, the effects of digital savings on emergency financing can be understood first through the reduction of borrowing, transfer and financial transaction costs. Secondly, the utilisation of digital savings enables the circumvention of constraints related to liquidity, time and mobility that may arise in emergency situations. However, this study reveals the heterogeneous impact of digital savings across regional economic communities and income levels, demonstrating that technological adoption, infrastructure development, and demographic composition jointly shape households' capacity to use digital savings effectively for emergencies.

To achieve the objective of this study, data concerning emergencies and digital savings has been sourced from the Global Findex for the years 2021, 2017, 2014 and 2011 for 42 SSA countries. The control variables have been taken from the World Bank's World Development Indicators. We then utilise a fractional Probit model with instrumental variables to assess the impact of digital savings on emergency financing, given that the variables measuring emergencies are proportions. The findings indicate that, with the exception of temporary deposits, all digital savings indicators exert a positive and significant influence on emergency financing for education and health in SSA. Furthermore, it has been demonstrated that digital savings fulfil the transaction and precautionary functions of traditional currencies. In addition, the effects of precautionary digital savings indicators appear to be more pronounced than those of transaction digital savings. The findings can be elucidated by the notion that digital savings empower households to circumvent numerous constraints in emergency circumstances, thereby reducing the financial burden associated with transactions, as postulated by constraint and transaction cost theories (Goldratt and Cox 1984; Williamson 1990).

This study shows that the size of the adult population has a significant impact on the probability of emergency financing. In addition, it is observed that final consumption, the share of credit to the private sector, the gender equality index, GDP per capita, and natural resource rents have a negative correlation with the probability of emergency financing. Furthermore, robustness analyses demonstrate that digital savings have a significant impact on emergency school financing as a country's level of development increases. Nevertheless, the process of development has the capacity to establish mechanisms that serve to restrict the utilisation of digital savings through avenues such as insurance, straightforward access to credit, and other comparable mechanisms. These mechanisms frequently serve to attenuate the impact of digital savings on emergency financing. However, it appears that the more developed a country's technological and institutional environment is, the more digital savings become a lever for financial inclusion and resilience. Finally, digital savings have been observed to influence emergency financing in a divergent manner across regional economic communities. The EAC and COMESA have demonstrated robust outcomes, while the SADC and ECCAS have exhibited a more varied set of results.

These results have important implications for policy and financial inclusion, indicating that expanding access to digital savings, improving digital literacy, and targeting marginalized populations can significantly strengthen household financial resilience, reduce reliance on high-cost informal borrowing, and enhance socio-economic development across Sub-Saharan Africa. By systematically linking digital savings utilization to both structural and behavioral factors, this study provides novel insights into how digital financial innovations can be leveraged to promote equitable and sustainable household financial security. The findings of this article initially support policies and programmes promoting the adoption and use of digital savings tools in SSA, as these generally facilitate the financing of emergencies. Such programmes may involve the reduction of digital tools, training and awareness-raising on digital tools, the development of digital infrastructure, and many other measures. Secondly, the results of the study argue for improving the financial capacity and purchasing power of households so that they can build up precautionary digital savings, which have a greater impact than transactional digital savings. Finally, in the interests of intra-country regional equity, efforts to develop and promote digital savings must be stepped up in rural areas, which generally suffer from low levels of financial and digital inclusion. In the interests of inter-country regional equity, supranational organisations should focus more on regional economic communities that are lagging behind in terms of digital savings and infrastructure through advice, project financing, training, regulation and other means. However, it is important to note that the findings of this article may be subject to re-evaluation in light of the emergence of novel digital savings tools such as artificial intelligence, big data, and data mining.

Appendix

See Appendix Table 7.

Table 7 List of countries

Angola	Congo, Dem. Rep	Guinea	Mauritius	Somalia
Benin	Congo, Rep	Kenya	Mozambique	South Africa
Botswana	Cote d'Ivoire	Lesotho	Namibia	South Sudan
Burkina Faso	Eswatini	Liberia	Niger	Sudan
Burundi	Ethiopia	Madagascar	Nigeria	Tanzania
Cameroon	Gabon	Malawi	Rwanda	Togo
Central African Republic	Gambia	Mali	Senegal	Uganda
Chad	Ghana	Mauritania	Sierra Leone	Zambia
Comoros				Zimbabwe

Sources: Authors, with Global Findex data

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